

Role of hydrogen fuel cells in achieving road transport decarbonization: A case study from Taiwan

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ABSTRACT

This study assesses the integration of hydrogen fuel cell vehicles (FCVs) with battery electric vehicles (BEVs) to support Taiwan's 2050 net-zero emissions goal. Using life cycle assessments and vehicle fleet modeling, we analyze hydrogen production pathways and forecast greenhouse gas (GHG) emissions across Taiwan's on-road vehicle fleet. Despite the BEV-FCV integration scenario, we project that the road transport sector could still emit 7.83 million tonnes of GHGs by 2050—145% above the government's target. Significant challenges include inadequate policies for electrifying heavy-duty vehicles, expected to contribute 50.4% of direct emissions in 2050, and the delayed phase-out of inefficient light-duty vehicles. Transitioning heavy-duty vehicles to zero-emission alternatives could reduce emissions by around 3.95 million tonnes. Our findings highlight the essential role of FCVs and robust hydrogen technologies in achieving sustainable transport, emphasizing the need for comprehensive electrification strategies.

Abbreviation and Nomenclature

ATR	auto-thermal reforming	kWh	kilowatt-hour
BEV	battery electric vehicle	LCA	life cycle assessment
BG	biomass gasification	LDT	light-duty truck
BR	biomass reforming	LDV	light-duty vehicle
CCS	carbon capture and storage	LH ₂	liquid hydrogen
CG	coal gasification	MCH	methylcyclohexane
CI	confidence interval	MT	methanol
CLHG	chemical looping hydrogen generation	NH ₃	ammonia
COGS	coke oven gas separation	PEMEC	proton exchange membrane electrolysis cell
DF	dark fermentation	RE	renewable energy
ER	energy recovery system	SMR	steam methane reforming
ET	ethanol	SOEC	solid oxide electrolysis cell
FCV	fuel cell vehicle	SR	survival rate
GH ₂	gaseous hydrogen	TTW	tank-to-wheel
GHG	greenhouse gas	TW-FLEET	Taiwan vehicle fleet model
GTT	gate-to-tank	VKT	vehicle kilometers traveled
HDT	heavy-duty truck	WTG	well-to-gate
HDV	heavy-duty vehicle	WTT	well-to-tank
HEV	hybrid electric vehicle	WTW	well-to-wheel

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HR	hydrogen recovery system	ZEV	zero emission vehicle
ICEV	internal combustion engine vehicle		

1. Introduction

Hydrogen energy offers a promising path toward revolutionizing energy systems and achieving decarbonization [1], due to its potential to replace fossil fuels which dominate global electricity generation [2] and contribute significantly to air pollution and greenhouse gas (GHG) emissions [3]. While hydrogen produces only water and heat during electricity and power generation, its production lifecycle, including source extraction, processing, and transportation, can still emit significant levels of carbon emissions [4]. Hydrogen production is classified into three types: grey hydrogen is produced from fossil fuels using technologies such as steam methane reformation [5]; blue hydrogen follows similar processes but incorporates carbon capture and storage (CCS) to reduce emissions [6]; and green hydrogen, produced via electrolysis using renewable energy, is recognized for its crucial role in

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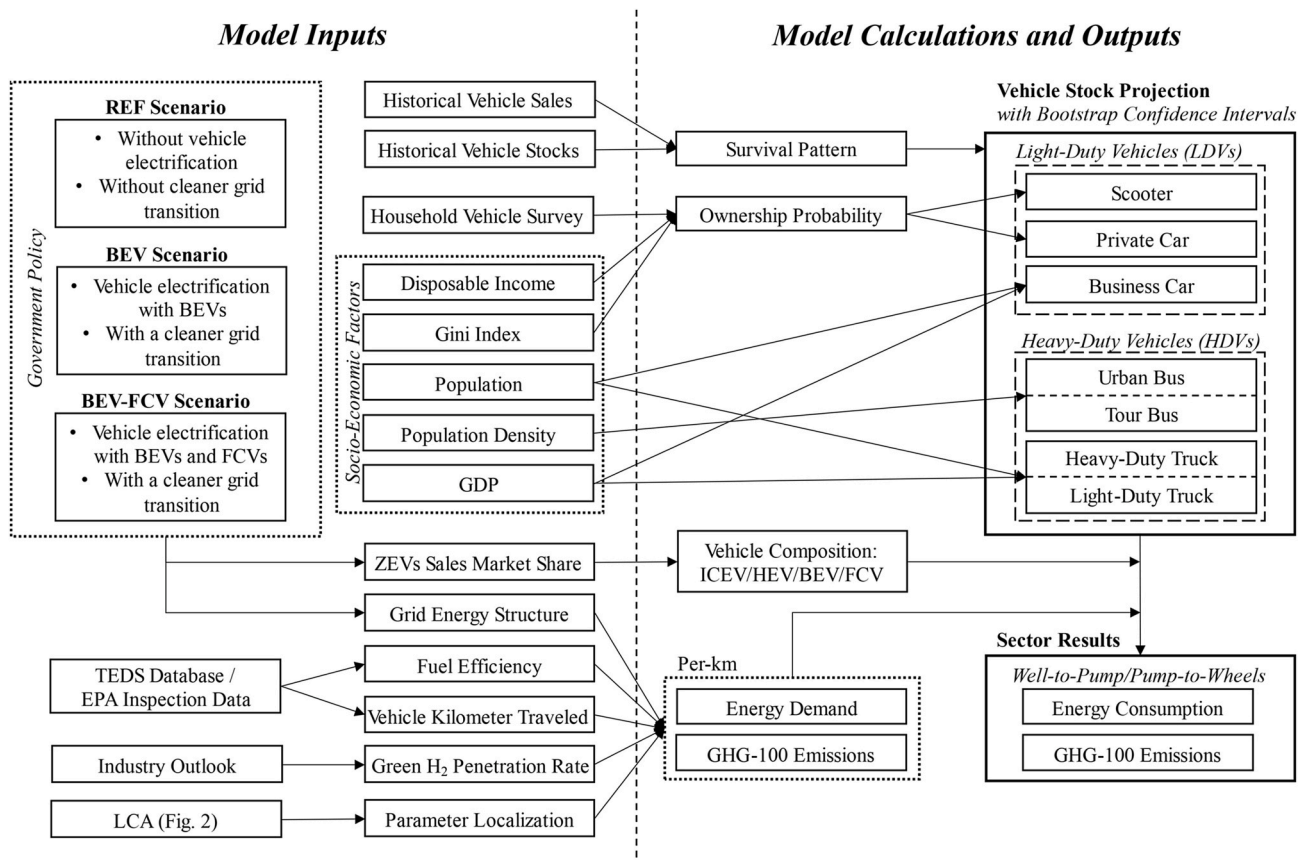


Fig. 1. Methodology and framework of this study.

global decarbonization efforts [7].

Global hydrogen production remains predominantly carbon-intensive, underscoring the urgent need for significant investment in low-carbon hydrogen infrastructure and research [8]. [9] suggested using fossil resources as a temporary solution for hydrogen production but emphasized the need for high-efficiency, low-emission processes. [10] analyzed five hydrogen production methods and found renewable energy methods less capable in terms of production capacity. [11,12] explored various technologies, notably highlighting the efficiency of chemical looping reforming and biomass-to-hydrogen processes for achieving low or negative GHG emissions. [13] conducted a comprehensive life cycle performance evaluation of diverse hydrogen technologies, stressing the importance of water consumption considerations in sustainability. [14] demonstrated that proton exchange membrane water electrolysis could cut CO₂ emissions by up to 75% with renewable energy. Finally [15], compared three blue hydrogen technologies, noting that auto-thermal reforming with CCS had the lowest life cycle GHG emissions at 3.91 kgCO₂eq/kgH₂. These studies collectively underscore the complex challenges and potential of scaling hydrogen production for environmental benefits.

Fuel cell vehicles (FCVs), classified as zero-emission vehicles (ZEVs), are recognized for their potential to decarbonize transportation [16]. FCVs are powered by electricity generated from an electrochemical process that combines hydrogen fuel, stored onboard in high-pressure tanks, with atmospheric oxygen. Notably, FCVs offer extended driving ranges and can be refueled in minutes, comparable in convenience to traditional internal combustion engine vehicles (ICEVs) [17]. Despite their potential, the adoption of FCVs is challenged by nascent hydrogen infrastructure and limited vehicle penetration [18]. As of 2022, there were only 72,000 FCVs globally compared to 26 million battery electric vehicles (BEVs), illustrating BEVs' dominance in the market [19]. Predominantly deployed in countries like South Korea and the USA, FCVs

are poised to significantly contribute to sustainable and low-carbon mobility alongside BEVs.

The carbon footprint of hydrogen vehicles, significantly influenced by production methods and fuel efficiency, is key to evaluating their environmental performance. [20] highlighted that hydrogen vehicles need superior efficiency (25–30% higher than gasoline-powered vehicles) to be competitive, especially when hydrogen is derived from natural gas. Utilizing renewable sources like wind for hydrogen production via electrolysis significantly lowers GHG emissions, suggesting a sustainable alternative [21]. Comparative studies, such as [22], have shown that hydrogen buses could markedly reduce urban GHG emissions by converting all urban buses to hydrogen fuel cell types, decreasing emissions by 0.2 MtCO₂eq annually. [23] emphasized the need for clear methodologies in comparing vehicle emissions, particularly between BEVs and FCVs. Lastly, [24] stressed the potential of hydrogen from renewable sources in reducing emissions from heavy-duty trucks, albeit cautioning against premature FCV adoption due to current grid dependencies. These studies collectively underscore the strategic importance of adopting hydrogen vehicles alongside robust lifecycle analysis to achieve significant environmental benefits.

In response to global climate change, Taiwan aims to become a net-zero emission society by 2050, targeting a drastic reduction in GHG emissions from road transportation—from 35.78 MtCO₂eq in 2019 to 3.19 MtCO₂eq [25]. As transportation is the third-largest source of GHG emissions in Taiwan, the challenge is substantial. The “Taiwan’s Pathway to Net-Zero Emissions in 2050 (the 2050 Net-Zero Pathway)” [26] outlines key BEV deployment milestones, including electrifying all urban buses by 2030 and banning the sale of fossil-fuel-powered scooters and private cars by 2040. However, Tsai et al. [27] predicted that existing electrification plans will be insufficient, projecting emissions of 8.4 MtCO₂eq by 2050, thus missing the target by 5.2 MtCO₂eq, partly due to inadequate planning for FCV deployment.

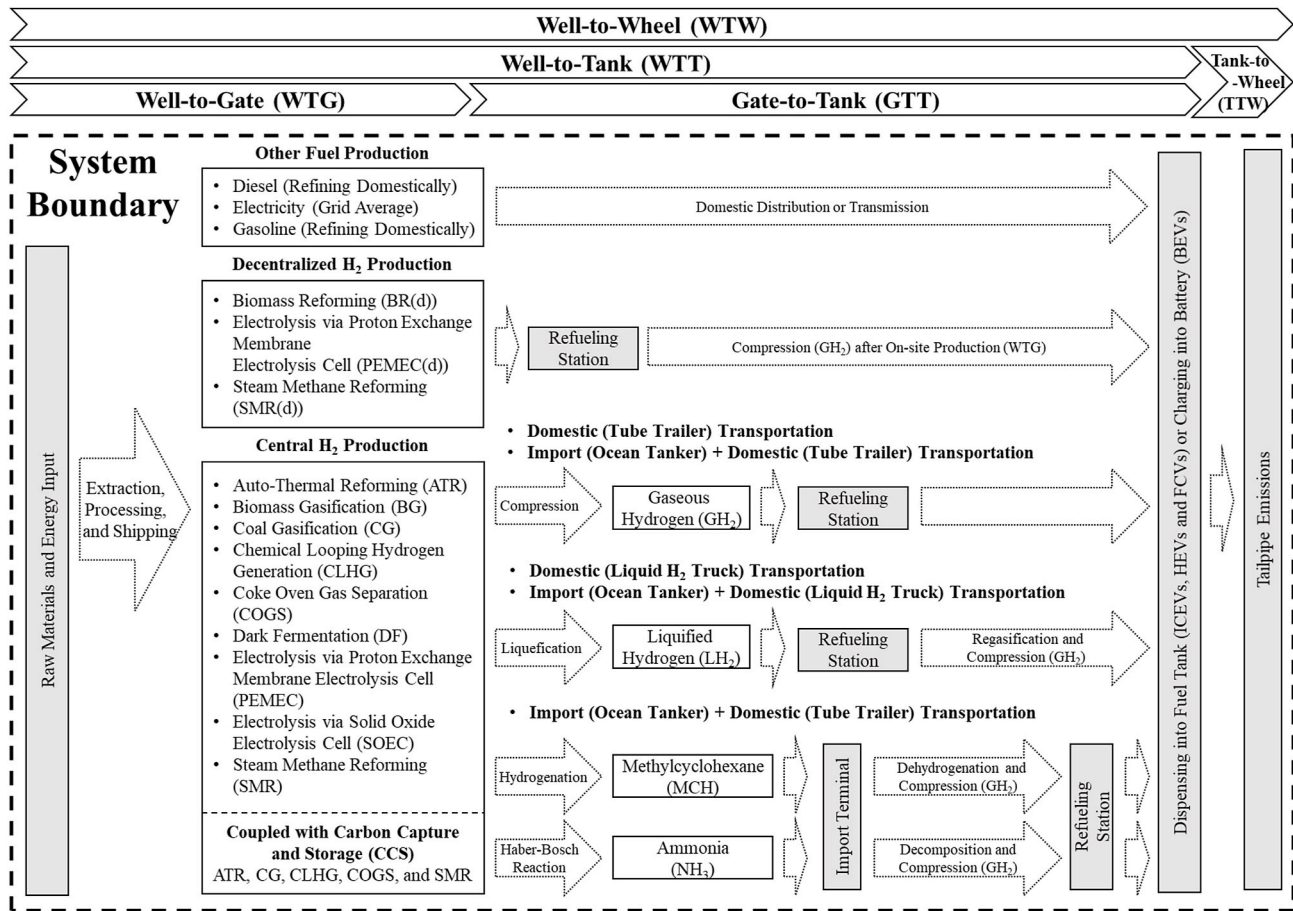


Fig. 2. Definition of system boundary for well-to-gate (WTG), gate-to-tank (GTT), and tank-to-wheel (TTW).

To enhance decarbonization efforts in the transport sector, this study employs the GREET model combined with local data for a comprehensive life cycle assessment of hydrogen FCVs. The refined and updated Taiwan Vehicle Fleet Model (TW-FLEET) assesses the socio-economic impacts on vehicle turnover and fleet composition, projecting energy use and emissions until 2050 across different ZEV technology deployment scenarios. This analysis seeks effective policy strategies to meet Taiwan's ambitious goals, highlighting the crucial roles of FCVs and BEVs and emphasizing the need to decarbonize heavy-duty vehicles in addition to light-duty fleets to significantly reduce carbon emissions and promote sustainable transportation.

2. Methodology and data

2.1. Research framework overview

Fig. 1 outlines the TW-FLEET framework used to evaluate long-term energy consumption and GHG emissions from Taiwan's on-road vehicle fleet until 2050. The LCA component (Section 2.2) focuses on diverse hydrogen production processes for mobility applications, enabling a thorough assessment of energy use and GHG intensities. In vehicle fleet turnover modeling (Section 2.3), socio-economic factors like disposable income and the Gini index are integrated to accurately capture fleet dynamics. To mitigate uncertainty in fleet forecasts, the bootstrap method is used for robust stock predictions, constructing confidence intervals through resampling [28]. Integrating LCA with fleet turnover modeling within TW-FLEET offers a holistic approach to analyzing vehicle fleet dynamics under various ZEV deployment scenarios (Section 2.4). Building on [27], this study enhances the TW-FLEET model by evaluating the environmental life cycle performances of diverse vehicle

powertrains. This expansion includes not only hydrogen FCVs but also hybrid electric vehicles (HEVs).¹ Furthermore, all model parameters have been carefully revised and updated to guarantee the accuracy and relevance of our analyses.

2.2. Life cycle assessment (LCA)

2.2.1. System boundary definition and assumptions

LCA evaluates the environmental impacts of vehicle production and fuel usage [30], adhering to the ISO 14040 and ISO 14044 standards which outline LCA principles and guidelines [31,32]. This study focuses on the fuel cycle energy consumption and carbon footprint of on-road vehicles powered by gasoline, diesel, electricity, and hydrogen in Taiwan. An entire LCA for the fuel cycle can be divided into three distinct stages (Fig. 2): well-to-gate (WTG), gate-to-tank (GTT), and tank-to-wheel (TTW) [33].

In the WTG stage, we consider the production of pure gaseous hydrogen, typically generated at central plants and distributed to refueling stations. The production of blue hydrogen in this stage integrates carbon capture via CCS, excluding the subsequent CO₂ treatment or storage. At the GTT stage, hydrogen is prepared for transport either as high-pressure gas or liquefied hydrogen using either grid power or renewable sources, with the study excluding losses during pressurization and storage. Transportation assumptions include domestic transfers via

¹ HEVs operate with a dual system of an internal combustion engine and an electric motor. The electric motor is powered by a battery, which is recharged via the engine or through regenerative braking, rather than being plugged into the grid [29].

high-pressure tube trailers or liquid hydrogen trucks and international imports in compound forms, decomposed back to hydrogen at ports. At refueling stations, hydrogen is stored and pressurized using local grid electricity, catering to the needs of FCVs filled at high pressures, typically 350 bar. In the TTW stage, the study focuses on hydrogen's conversion into power through vehicles' engines, motors, or fuel cells, considering the overall WTW impacts, which represent the total environmental effects of the fuel across its lifecycle [34].

$$WTT_i = WTG_i + GTT_i \quad (1)$$

$$WTW_i = WTG_i + GTT_i + TTW_i \quad (2)$$

Where.

WTT_i = Well-to-tank environmental impact in year i , referring to aggregate emissions prior to end-use

WTG_i = Well-to-gate environmental impact in year i , referring to upstream and manufacturing emissions

GTT_i = Well-to-tank environmental impact in year i , referring to emissions from transport and treatment to facilitate transport

WTW_i = Well-to-wheel environmental impact in year i , referring to overall fuel cycle emissions

TTW_i = Tank-to-wheel environmental impact in year i , referring to emissions from vehicles in operation

2.2.2. Parameters localization of the GREET model

The Greenhouse Gases, Regulated Emissions, and Energy Use in Transportation (GREET) model, developed by Argonne National Laboratory and sponsored by the U.S. Department of Energy [35], serves as an essential LCA tool. To ensure the GREET model's relevance to Taiwan, parameters are localized to reflect the region's unique energy dynamics, particularly its reliance on imported energy due to geographic constraints [36]. Adjustments are made to elements like the origin of primary energy sources, transport distances and methods, the structure of power generation, and average vehicle mileage. Details on these parameter modifications are available in the [Supplementary Information Section 1.1](#).

2.2.3. Hydrogen production pathway

This study assesses various hydrogen production methods through literature reviews and utilizes the hydrogen LCA pathways defined in the GREET model. It explores hydrogen production from fossil fuels, biomass, and water electrolysis. Fossil-fuel-based hydrogen is generated through thermochemical processes such as reforming or gasification [37–39], covering five technologies: auto-thermal reforming (ATR), coal gasification (CG), chemical looping hydrogen generation (CLHG), coke oven gas separation (COGS), and steam methane reforming (SMR). ATR burns part of the feedstock to produce the required heat, differing from SMR which needs external heating [40]. COGS recovers hydrogen from coke oven gas, a by-product of coal-to-metallurgical coke conversion in the steel industry [41,42]. CLHG employs metal oxide as an oxygen carrier, ensuring efficient purification of hydrogen and capture of carbon dioxide [43].

Three biomass-based methods—biomass gasification (BG), biomass reforming (BR) with ethanol or methanol, and dark fermentation (DF) integrated with energy or hydrogen recovery systems—are also considered. While BG and BR are categorized as thermochemical processes [44], DF represents a biological process that ferments organic matter to produce biohydrogen in a light-free environment [45]. Additionally, proton exchange membrane electrolysis cell (PEMEC) and solid oxide electrolysis cell (SOEC) systems are evaluated. These utilize electricity to split water into hydrogen and oxygen, with the electricity sourced from either the grid or renewable energies like solar or wind [46]. Comprehensive details on these processes, including life cycle inventory sources and assumptions for well-to-tank processes, are detailed

in the [Supplementary Information Section 1.2](#).

2.2.4. Hydrogen transportation pathway

Hydrogen transportation involves transporting diatomic hydrogen in either gaseous (GH₂) or liquid (LH₂) forms. GH₂ is compressed to high pressures, typically between 250 and 700 bar [47], whereas LH₂ is cooled to cryogenic temperatures below −244 °C [48]. GH₂ requires larger tanks or more frequent transport due to its low density, while LH₂ demands higher costs for cryogenic containment [49]. Both methods necessitate specialized infrastructure due to their storage and handling requirements [50]. Alternatively, hydrogen can be transported as compounds like ammonia (NH₃) or methylcyclohexane (MCH) [51], offering higher volumetric density and less stringent transport conditions [52]. MCH is produced by hydrogenating toluene to yield MCH and is dehydrogenated upon arrival to release hydrogen [53]. NH₃ is produced via the Haber-Bosch method by reacting hydrogen with nitrogen [54] and decomposing back to hydrogen at the point of use [55]. The study also explores decentralized hydrogen production (or distributed, denoted as (d)), where hydrogen is generated and used on-site at refueling stations, eliminating the need for long-distance transport. For further details on the hydrogen transport pathways, see [Supplementary Information Section 1.3](#).

2.2.5. Energy consumption and GHG emissions

The energy consumption (demand) of road transport is influenced by several factors, including the number of vehicles in stock, fuel type, fuel efficiency, vehicle kilometer traveled (VKT), and upstream fuel cycle emissions. In the GREET algorithm, the energy consumption of fuels is the aggregate of the energy consumption of all upstream feedstocks and the energy consumption of the processes or technologies used to manufacture fuels. The corresponding GHG emissions are the multiplication of energy consumption and GHG emission intensity. The total energy consumption and GHG emissions in the road transport sector are calculated using the formulas shown in Equations (3) and (4). Fuel type, fuel efficiency, and VKT data are sourced from established databases, with specific details for FCVs included. For comprehensive metrics used in this analysis, see [Supplementary Information Section 1.4](#).

$$EC_{s,i} = \sum_p \sum_t V_{t,p,i} \cdot S_{t,p,i} \cdot FE_{t,p,i}^{-1} \cdot VKT_{t,p,i} \cdot \sum_h EC_{h,s,p,i} \quad (3)$$

$$GHG_{s,i} = \sum_p \sum_t V_{t,p,i} \cdot S_{t,p,i} \cdot FE_{t,p,i}^{-1} \cdot VKT_{t,p,i} \cdot \sum_h (EC_{h,s,p,i} \cdot EI_{h,s,p,i} + GHG_{other,h,s,p,i}) \quad (4)$$

Where.

$EC_{h,s,p,i}$ = Energy consumption in process h in life cycle stage s when producing fuel or energy for use in powertrain type p in year i (kWh)

$V_{t,p,i}$ = Stock of vehicle type t with powertrain type p in year i (thousands or millions of vehicles)

$S_{t,p,i}$ = Stock share of a vehicle type t with powertrain type p in year i (%)

$FE_{t,p,i}$ = Fuel efficiency of a vehicle type t with powertrain type p in year i (km/L for ICEV and HEVs; km/kWh for BEV; km/kgH₂ for FCV)

$VKT_{t,p,i}$ = Annual vehicle kilometers traveled for vehicle type t with powertrain type p in year i (km)

$GHG_{s,i}$ = GHG emissions in life cycle stage s in year i (gCO₂eq or MtCO₂eq)

$EI_{h,s,p,i}$ = GHG emissions intensity (per energy consumption) in the process h in life cycle stage s when producing fuel or energy for use in powertrain type p in year i (gCO₂eq/kWh)

GHG_{other} = GHG emissions that are not attributed to energy-consuming processes or technology, i.e., leakage, fugitive or transmission loss (gCO₂eq)

2.3. Vehicle fleet model

2.3.1. Vehicle ownership and stock

The road vehicles in Taiwan are classified into five primary categories, taking into account their stock growth rates, usage intensity, and potential for electrification: scooters, private cars, business cars (including rental cars and taxis), buses (including urban and tour buses), and freight vehicles (including heavy-duty trucks (HDTs) and light-duty trucks (LDTs)). Each module specific to vehicle types considers a range of socio-economic factors, such as income, Gini index, population density, population size, and gross domestic product (GDP), to estimate the stocks within each category. Ownership patterns vary with income; for scooters, the relationship is an inverted U-shape (Equation (5)), suggesting a peak in ownership at mid-income levels. In contrast, car ownership increases in an S-shaped curve (Equation (6)), indicating growth with higher incomes.² Vehicle sales are segmented into new purchases driven by rising incomes and replacements for aging vehicles, reflecting the auto market's maturity. This study updates these ownership functions (Equations (7)–(12)) [27] with the most recent demographic data to provide updated projections of on-road vehicle turnover until the year 2050.

$$f(x) = a \bullet \exp(b \bullet \exp(c \bullet x)) \quad (5)$$

$$g(x) = u^v \bullet \frac{x^{v-1} \exp(-u \bullet x)}{\Gamma(v)} + w \quad (6)$$

$$I_i(x_{1,i}, z(x_{2,i})) = \frac{\left(\frac{x_{1,i}}{w}\right)^{z-1}}{\left(1 + \left(\frac{x_{1,i}}{w}\right)^z\right)^2} \quad (7)$$

$$V_{scooter,i} = x_{3,i} \int_0^\infty I_i(x_{1,i}) f(x_{1,i}) dx_1 \quad (8)$$

$$V_{private\ car,i} = x_{3,i} \int_0^\infty I_i(x_{1,i}) g(x_{1,i}) dx_1 \quad (9)$$

$$V_{business\ car,i} = p \exp(q \exp(rx_{4,i})) \quad (10)$$

$$V_{bus,i} = \frac{\zeta}{(1 + \exp(\eta(x_{5,i} - \theta)))} + \kappa(i - 1998) \quad (11)$$

$$V_{freight,i} = \tau x_{3,i} + \phi \ln(x_{4,i}) + \psi \quad (12)$$

Where,

$x_{sf,i}$ = Socio-economic factor sf in year i ; x_1 is the disposable income per capita (million NT\$), x_2 is the Gini index, x_3 is population (people), x_4 is gross domestic product (GDP) per capita (million NT\$ per capita), x_5 is population density (people per square kilometer)
 f = Modified gamma distribution, an inverted U-shaped relationship for scooter ownership; $(\alpha, \beta, \gamma) = (1.48, 0.96, 0.17)$.
 g = Gompertz function, an S-shaped relationship for car ownership; $(a, b, c) = (0.64, -2.00, -2.42)$.
 I_i = Disposable income distribution with log-logistic distribution assumption in year i (%); Values for $(w, z) = (0.196, 3.067)$ in 2000, $(0.237, 2.924)$ in 2010, $(0.293, 2.941)$ in 2020, $(0.345, 2.933)$ in

2030, $(0.397, 2.933)$ in 2040, and $(0.450, 2.933)$ in 2050 $V_{scooter,i}$ = Stock of scooters in year i (millions of vehicles)

$V_{private\ car,i}$ = Stock of private cars in year i (millions of vehicles)

$V_{business\ car,i}$ = Stock of business cars in year i (thousands of vehicles); $(p, q, r) = (0.25, -230.06, -9.61)$.

$V_{bus,i}$ = Stock of buses in year i (thousands of vehicles); $(\zeta, \eta, \theta, \kappa) = (2.38, -0.73, 2.18, 0.017)$.

$V_{freight,i}$ = Stock of freight vehicles in year i (millions of vehicles); $(\tau, \phi, \psi) = (0.081, 2.85 \times 10^5, -4.69 \times 10^6)$.

2.3.2. Vehicle survival rate

Vehicle survival patterns are essential for predicting the turnover of on-road fleets. The survival rate (SR) indicates how likely a vehicle is to remain operational over time [56,57]. For scooters and private cars, replacements mainly occur as vehicles reach their life's end, modeled using a two-parameter logistic function to represent age-dependent attrition (Equation (13)). Business cars follow a Weibull distribution for survival (Equation (14)), though discrepancies between model predictions and actual early-life data lead to using real first-year survival rates in the study. For buses, survival aligns with regulated service life limits, applying a direct assignment method to ensure no premature decommissions (Equation (15)). The decay in the operational likelihood of heavy and light-duty trucks is modeled with an exponential function, reflecting their decreasing survival rates with aging (Equation (16)).

$$SR_1(k) = \frac{1}{(1 + \exp(nk))} \quad (13)$$

$$SR_2(k) = \exp\left(\left(\frac{-k}{d}\right)^q\right) \quad (14)$$

$$SR_3(k) = \begin{cases} 1, & k \leq y \\ 0, & k > y \end{cases} \quad (15)$$

$$SR_4(k) = \exp(-rk) \quad (16)$$

Where,

SR_1 = Survival rate for vehicle age k ; $(m, n) = (0.00084, 0.35)$ for private cars and $(0.0063, 0.29)$ for scooters

SR_2 = Survival rate for vehicle age k ; $(d, q) = (5.81, 1.24)$ for business cars

SR_3 = Survival rate for vehicle age k ; $y = 8$ for urban buses and $y = 15$ for tour buses

SR_4 = Survival rate for vehicle age k ; $r = 0.0602$ for HDTs and $r = 0.0362$ for LDTs

2.4. Scenario design for BEV and FCV development

This study assesses the impact of government policies on ZEV market penetration through three distinct scenarios, with comparative sales market shares visualized in Fig. S11:

- (1) **Reference Scenario (REF scenario):** This scenario assumes cessation of current BEV promotion policies, projecting that most vehicles will remain gasoline or diesel-powered through 2050. Changes in road transport environmental burden primarily arise from market developments and fuel efficiency improvements. It serves as a baseline for comparing carbon reduction outcomes across scenarios, assuming constant carbon emission intensity from electricity generation post-2022.
- (2) **BEV Deployment Scenario (BEV scenario):** Aligned with the government's current policy framework and the 2050 Net-Zero Pathway, this scenario targets electrification of all urban buses by 2030 and all new car and scooter sales by 2040. Notably, tour buses and freight vehicles remain ICEVs and are excluded from

² The impact of car pricing on vehicle ownership is not included in our model due to the mature automotive market in Taiwan, where disposable income rather than vehicle affordability predominantly influences car ownership, as documented by Tsai et al. (2023) [27].

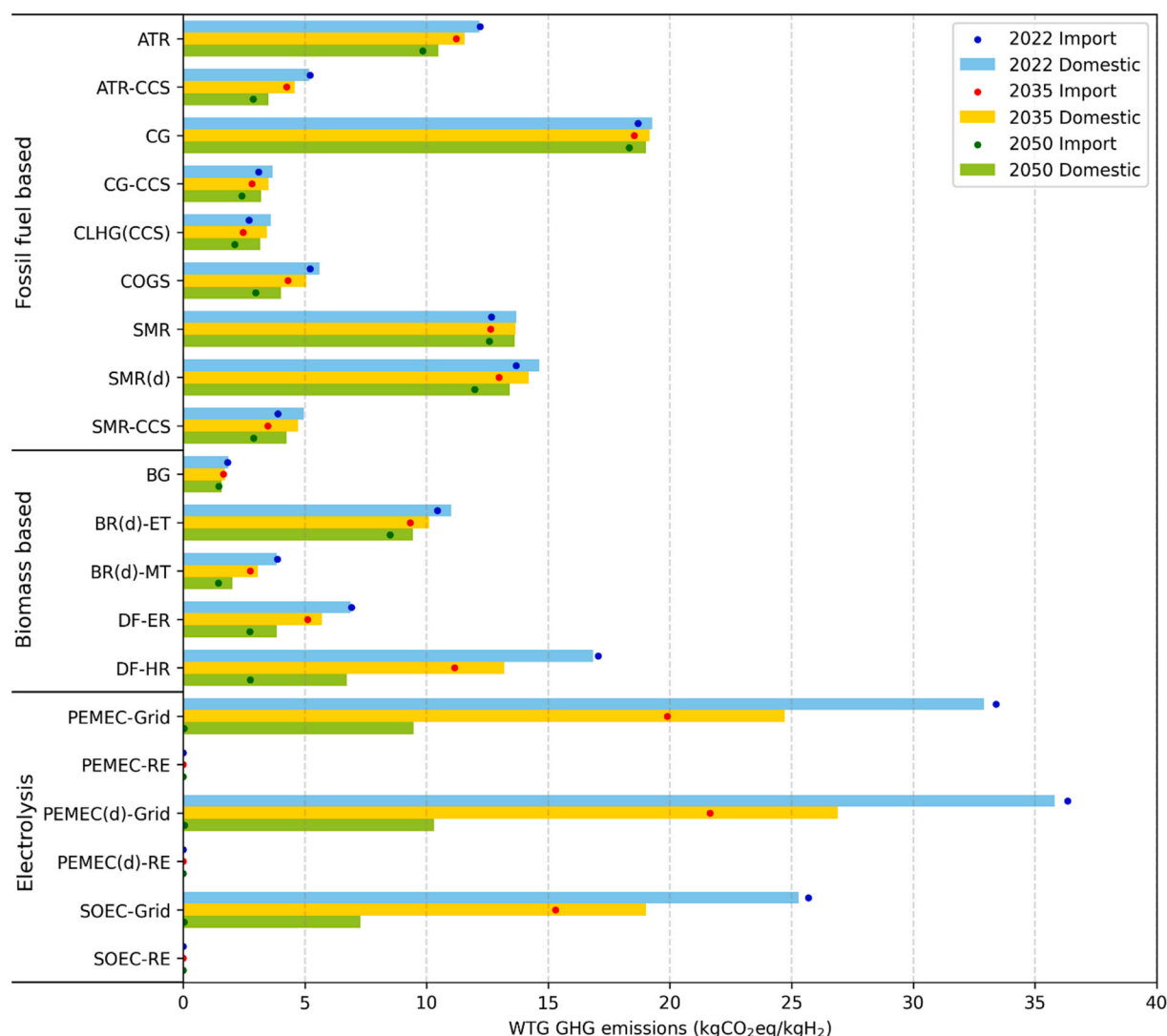


Fig. 3. Life cycle GHG emissions in the WTG stage of hydrogen production (2022, 2035, 2050).

electrification. Unlike the REF scenario, this scenario anticipates a decrease in grid emission intensity from now until 2050.

- (3) **BEV-FCV Scenario:** Building on the BEV scenario, this explores the potential of FCVs to complement BEVs in reducing road transport emissions, particularly for heavy-duty vehicles like tour buses and trucks, which are not fully covered in the BEV scenario. This scenario posits FCVs as zero-emission alternatives to ICEVs and HEVs, setting sales targets of 5% by 2030 and 30% by 2050 for LDVs, and 20% by 2030 and 50% by 2050 for HDVs, based on extrapolations from global policy objectives (Table S7). This aims to evaluate the efficacy of FCVs in significantly lowering GHG emissions and contributing to deep decarbonization goals.

3. Results and discussion

3.1. Hydrogen production (well-to-gate) emissions comparison

Fig. 3 presents the LCA results of GHG emissions for the WTG stage of hydrogen production in Taiwan, highlighting the significant impact of the carbon intensity of Taiwan's grid and the potential for mitigating these emissions through future improvements in grid carbon footprint and production technologies. The energy consumption details are provided in Supplementary Information Section 1.5.

For hydrogen production relying on fossil fuels, disparities in GHG

emissions are evident. CG shows the highest carbon footprint at 19.3 kgCO₂eq/kgH₂ in 2022, with only slight reductions by 2050. SMR and ATR follow, with emissions of 13.7 and 12.2 kgCO₂eq/kgH₂, respectively. The ATR process is expected to see a 13.7% reduction by 2050, thanks to improved efficiency in oxygen integration from a grid-powered air separation unit. Implementing CCS significantly reduces the carbon footprint of these processes, with CG-CCS showing an 80.9% decrease, and SMR and ATR with CCS reducing emissions by 62.4% and 55.9%, respectively. CLHG sets itself apart by not requiring additional CCS [58], starting with a carbon footprint of 3.7 kgCO₂eq/kgH₂ in 2022 and decreasing slightly by 2050. The COGS process, which recovers hydrogen from the steel industry's by-products, has its own GHG emissions modeled within the GREET model's inventory, with a starting value of 5.6 kgCO₂eq/kgH₂ in 2022 and a forecasted 28.2% reduction by 2050.

When comparing thermochemical processes, BG is notable for its low carbon footprint, starting at 1.86 kgCO₂eq/kgH₂, and expected to reduce by 15.9% by 2050. This is in contrast to the BR process, which requires prior conversion to ethanol and methanol, leading to higher energy and material requirements and, consequently, higher GHG emissions. The emissions from DF processes are considerable, at 6.9 kgCO₂eq/kgH₂ for systems with energy recovery and 16.9 kgCO₂eq/kgH₂ for those with hydrogen recovery in 2022. These processes offer significant potential for emission reductions as Taiwan's grid becomes less carbon-intensive,

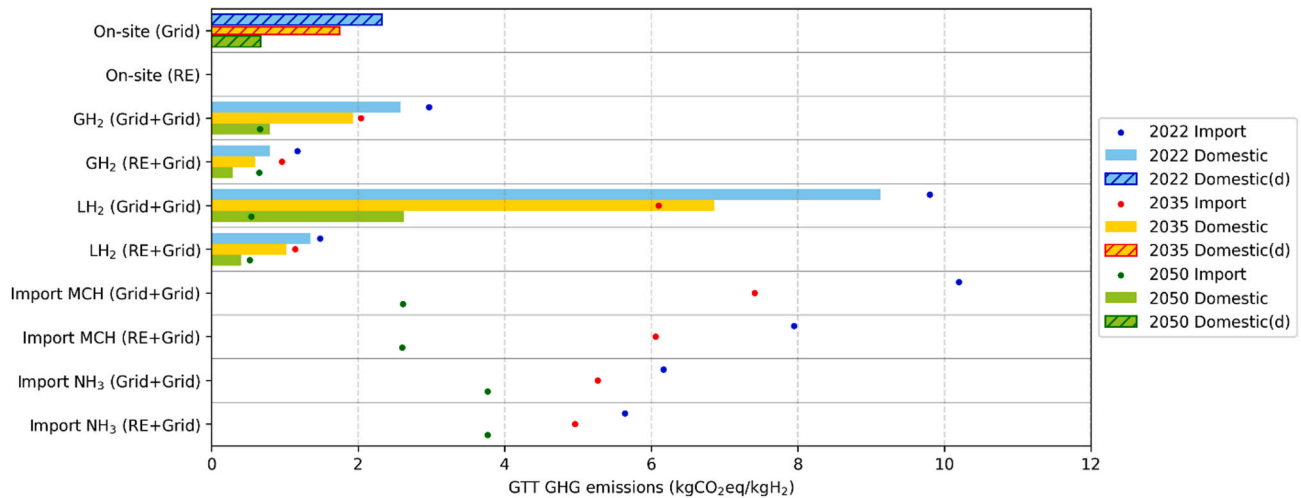


Fig. 4. Life cycle GHG emissions in the GTT stage of hydrogen transportation (2022, 2035, and 2050).

with projected decreases of 43.9% and 60.0% by 2050, respectively.

The electrolysis of water for hydrogen production also presents a significant opportunity for reducing GHG emissions, especially if there is a transition to a less carbon-intensive grid. Currently, the high carbon intensity of Taiwan's grid results in elevated emissions for domestically produced hydrogen, particularly in comparison to imported hydrogen from regions with greener energy practices like Australia (Supplementary Information Section 1.6). However, with projected grid improvements, the carbon footprint of hydrogen production by electrolysis could decrease by as much as 71.2% by 2050, potentially reaching zero emissions if powered entirely by renewable energy sources. This scenario aligns with global decarbonization goals, positioning electrolysis as a sustainable method for future hydrogen production.

3.2. Hydrogen transportation (gate-to-tank) emissions comparison

Fig. 4 showcases the GHG emissions during the transportation and dispensing of hydrogen, providing a detailed analysis of the GTT stage across various transport methods, both domestic and international. This focus aims to elucidate the environmental impacts specific to the technical aspects of hydrogen transport without delving into the equipment

specifications or the economic variables of transportation efficiency.

For the transport of GH₂ within Taiwan, the carbon emissions in 2022 range between 1.2 and 3.0 kgCO₂eq per kg of hydrogen, depending on the energy source for compression at central plants. Tube trailer shipment, a prevalent method for domestic transportation, exhibits a relatively minor carbon footprint of 0.16 kgCO₂eq per kgH₂. Conversely, transporting LH₂ generates significantly higher emissions of 9.8 kgCO₂eq per kgH₂ when utilizing grid electricity, reduced to 1.48 kgCO₂eq per kgH₂ with renewable energy inputs at the plant level. The liquefaction process, essential for LH₂ transport, is the major source of GHG emissions in this stage, primarily due to the intensive electricity demand. However, the condensed nature of LH₂ minimizes the carbon footprint per transported unit to just 0.03 kgCO₂eq per kgH₂. Storage, vaporization, and compression at refueling stations further contribute to emissions. For international routes, using ocean tankers lowers the carbon footprint to 0.372 kgCO₂eq for GH₂ and 0.119 kgCO₂eq for LH₂, highlighting the efficiency of shipping hydrogen in compact liquid or gaseous forms.

Transporting hydrogen compounds like NH₃ and MCH introduces additional complexities. NH₃ shows a lower carbon footprint than LH₂ when processed with grid electricity, whereas MCH has slightly higher

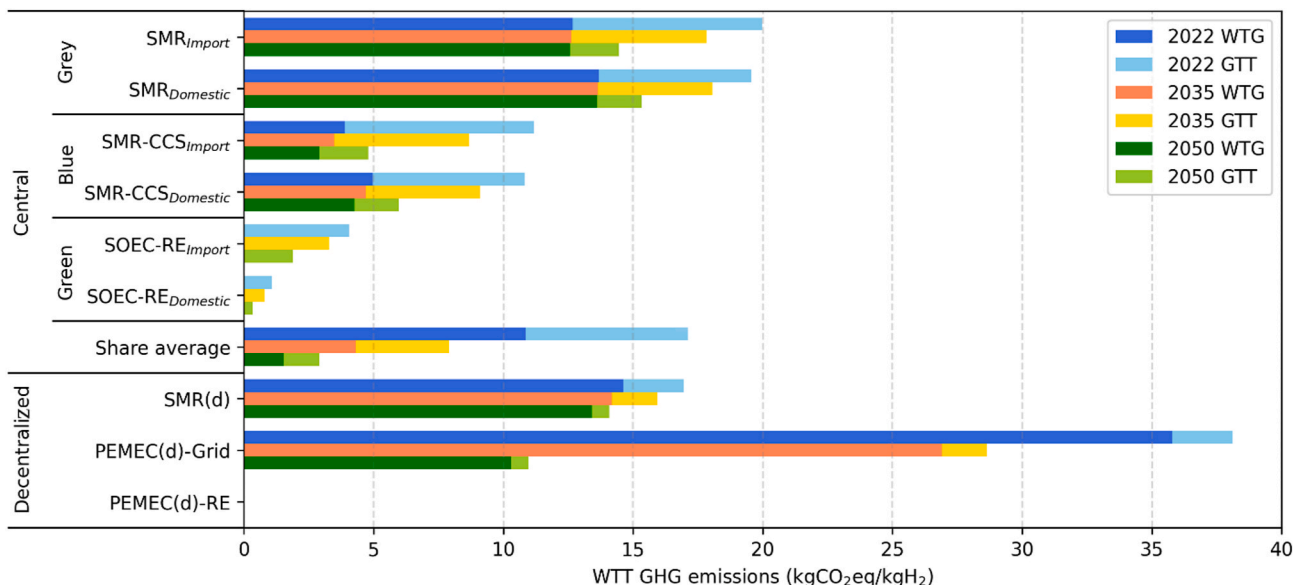


Fig. 5. Life cycle GHG emissions in the WTT stage of hydrogen production pathways and sources (2022, 2035, and 2050).

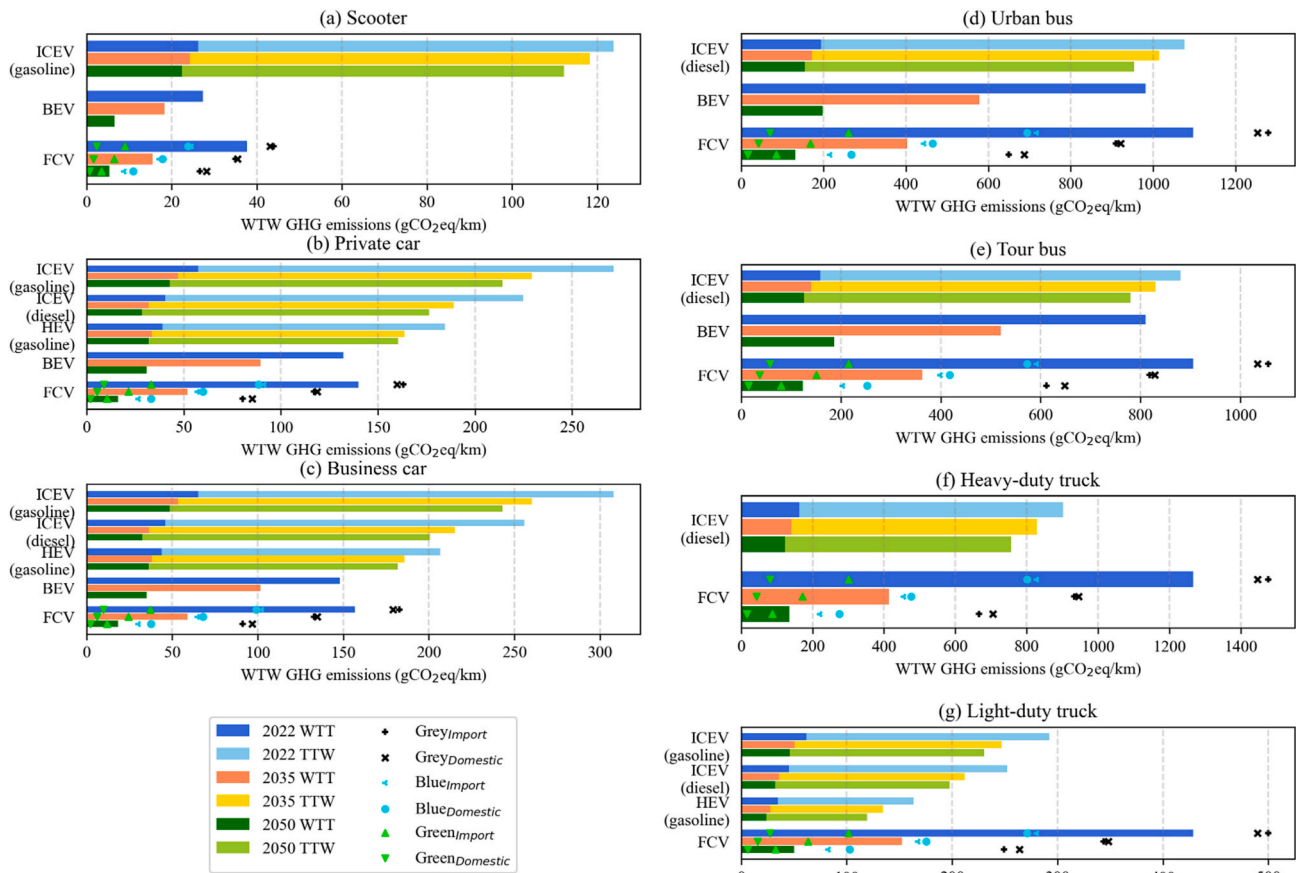


Fig. 6. WTW fuel life cycle GHG emissions by vehicle type and powertrain technology for on-road vehicles in Taiwan, spanning the years 2022, 2035, and 2050. LDVs: (a) scooter, (b) private car, (c) business car; HDVs: (d) urban bus, (e) tour bus, (f) heavy-duty truck, (g) light-duty truck.

emissions. As Taiwan's energy grid evolves towards greener sources, the emission advantages will shift towards LH₂ and then MCH, expected to surpass NH₃ by 2050. With Australia projected to achieve a nearly zero-carbon electrical grid by 2050—a significant source of hydrogen imports—transporting MCH or NH₃ might display comparable footprints whether using grid electricity or renewable energy. Ultimately, transporting GH₂ or LH₂ using renewable energy remains the most beneficial option due to the lack of additional synthesis processes required. The decision between domestic GH₂ transportation and imported hydrogen forms depends on the local grid's carbon intensity and the availability of renewable energy for liquefaction processes. Should renewable sources be readily available, imported LH₂ could present a lower carbon footprint than GH₂, emphasizing the advantages of moving hydrogen in more condensed forms under renewable energy conditions.

3.3. Hydrogen production pathways (well-to-tank) and source emissions comparison

This section integrates the WTG and GTT stages to evaluate the WTT GHG emissions from various hydrogen production pathways and sources. Fig. 5 presents a comparative analysis of WTT GHG emissions for selected hydrogen production methods over the years 2022, 2035, and 2050, highlighting the efficiency of domestic versus imported hydrogen and centralized versus decentralized production methods.

SMR is assessed as the standard for grey hydrogen due to its widespread industrial use [59]. Additionally, SMR combined with CCS technology is considered for blue hydrogen production. Green hydrogen is exemplified by SOEC utilizing renewable energy at the production site. The analysis also considers the weighted average market shares of these hydrogen types based on the International Energy Agency's

forecasts of global hydrogen-based fuel production (Fig. S2) [8], assuming an equal distribution between domestic production and imports.

The findings indicate that domestically produced hydrogen generally exhibits a higher WTT carbon footprint compared to imported hydrogen for 2022 and 2035, except for green hydrogen. This suggests that despite the proximity benefits of local production, the significant emissions from the WTG stage, driven by the local grid's carbon intensity, negate these advantages. For decentralized production methods, the carbon footprint depends on the energy source used for hydrogen production. Using grid electricity at refueling stations can lead to a higher carbon footprint than that of centrally produced grey hydrogen. In contrast, decentralized grey hydrogen production (SMR(d)) yields a lower WTT footprint as it eliminates transportation emissions but adds more at the production phase.

The most environmentally beneficial method involves producing hydrogen at refueling stations using renewable energy, achieving almost zero emissions throughout the hydrogen fuel cycle. Following closely, domestically produced green hydrogen shows commendable environmental performance, though its practical implementation is limited by the current state of Taiwan's hydrogen infrastructure and renewable energy capabilities. In scenarios where technological and infrastructural constraints persist, importing green or blue hydrogen appears as a favorable alternative to minimize the carbon footprint of the hydrogen fuel cycle. This approach leverages advancements in production technologies and existing global infrastructure to meet local energy needs sustainably.

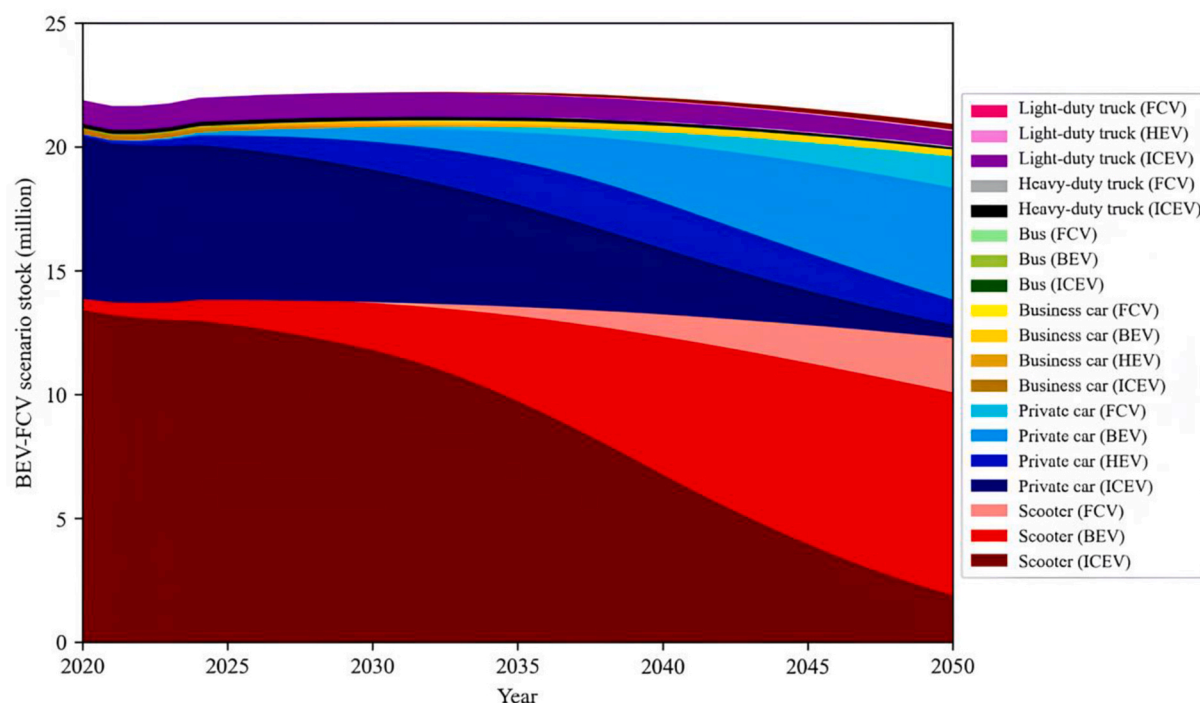


Fig. 7. Median forecast of on-road vehicle stocks in Taiwan: A BEV-FCV scenario perspective.

3.4. Comparative life cycle analysis of ICEVs, HEVs, BEVs, and FCVs

Fig. 6 illustrates the WTW GHG emissions across different vehicle powertrains from 2022 to 2050, showcasing significant variances in emissions reductions across vehicle types. In 2022, BEVs demonstrated substantial reductions in GHG emissions compared to ICEVs, HEVs, and FCVs among LDVs like scooters and cars. For instance, an electric private car was found to emit only 48.6% of the GHG per kilometer compared to its gasoline-powered counterpart. FCVs showed lower WTW emissions than BEVs only when using green or blue hydrogen as their fuel source.

By 2035, it is projected that FCVs will overtake BEVs in terms of lower fuel cycle carbon emissions, especially if the reliance on grey hydrogen drops to 20% as anticipated globally [8]—a shift that aligns with findings by [24]. For HDVs, including buses and trucks, the emission scenarios are complex. Currently, HDVs powered by FCVs could potentially emit more GHGs than ICEVs, contingent upon the hydrogen production method and vehicle fuel efficiency. Yet, with a projected decline in grey hydrogen use and expected improvements in fuel efficiency—foreseen to increase by 25–60% by 2050 according to the GREET model—FCVs are anticipated to increasingly rival BEVs in emissions efficiency.

The period leading to 2035 represents a pivotal phase for promoting FCV adoption, especially within the HDV category where advancements in vehicle technology could lead to notable GHG reductions. This analysis integrates emissions data from various vehicle types and powertrains using the TW-FLEET model, projecting total emissions for Taiwan's on-road transport sector. This approach establishes a solid base for devising strategic policies and initiatives aimed at achieving comprehensive decarbonization goals within the transport sector.

3.5. Evolution of vehicle stocks and penetration rates

This segment of the study builds upon the foundational analysis by [27], which assessed the environmental impact of BEVs under current policy frameworks. It explores the integration of low-carbon hydrogen FCVs into Taiwan's vehicle fleet, using the updated TW-FLEET model enriched with the latest socio-economic data and baseline FCV adoption

rates. Fig. 7 illustrates the forecasted median on-road vehicle stocks in the BEV-FCV scenario from 2022 to 2050. The model predicts a peak in vehicle numbers around 22.24 million by 2032 (95% CI: 21.72–22.42 million), followed by a gradual decline to 21.99 million vehicles by 2040 (95% CI: 21.31–22.23 million) and further down to 20.94 million by 2050 (95% CI: 20.00–21.27 million). This trajectory highlights the dynamic evolution of vehicle stocks, influenced by advancements in automotive technology and the strategic adoption of FCVs alongside BEVs, aiming for a sustainable and low-carbon transport sector. Detailed outcomes from the TW-FLEET model are available in the [Supplementary Information Section 2.2](#).

Projected adoption rates for FCVs by 2050 across various vehicle types are as follows: scooters at 17.7%, private cars at 16.9%, business cars at 24.5%, urban buses at 43.8%, tour buses at 38.8%, heavy-duty trucks at 29.0%, and light-duty trucks at 26.1%. Fig. 8 compares these forecasts with those of the REF and BEV scenarios, indicating that ZEVs could represent 79.6% of the on-road fleet by 2050, with BEVs accounting for 61.5% and FCVs 18.0%. This marks a substantial increase from the 8.58% in the REF scenario and 78.3% in the BEV-only scenario. Notably, the extra 1.3% in the BEV-FCV scenario comes from the inclusion of FCVs, particularly benefitting tour buses, heavy-duty trucks, and light-duty trucks, which are not currently covered by the government's electrification plans.

3.6. Fuel life cycle GHG emissions under different scenarios

Fig. 9 presents a comprehensive analysis of fuel life cycle GHG emissions across various scenarios, highlighting the diverse impacts of different vehicle types and powertrain technologies (see Fig. S16 for the associated energy consumption results). The results reveal the transition's initial surge in WTT emissions due to upstream electricity generation and hydrogen production, which contrasts with direct TTW emissions from conventional vehicle operations. Despite this early increase, projections show a decline in WTT emissions by 2042 as the integration of renewable energy sources progresses, reducing emissions to levels below those seen in the REF scenario. Concurrently, a significant decrease in TTW emissions results from the electrification of

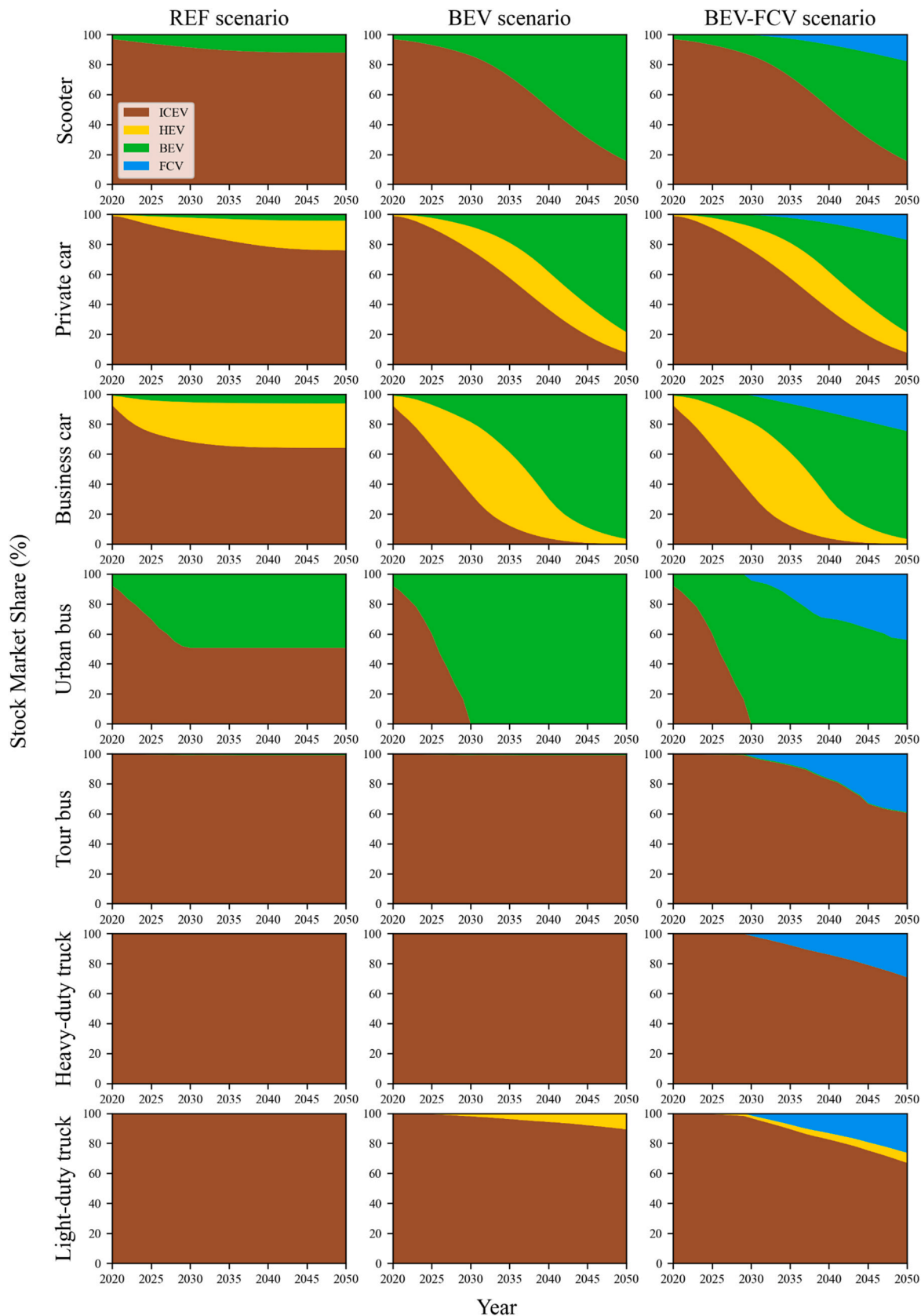


Fig. 8. Comparative on-road vehicle stock market share forecasts: REF, BEV, and BEV-FCV scenarios.

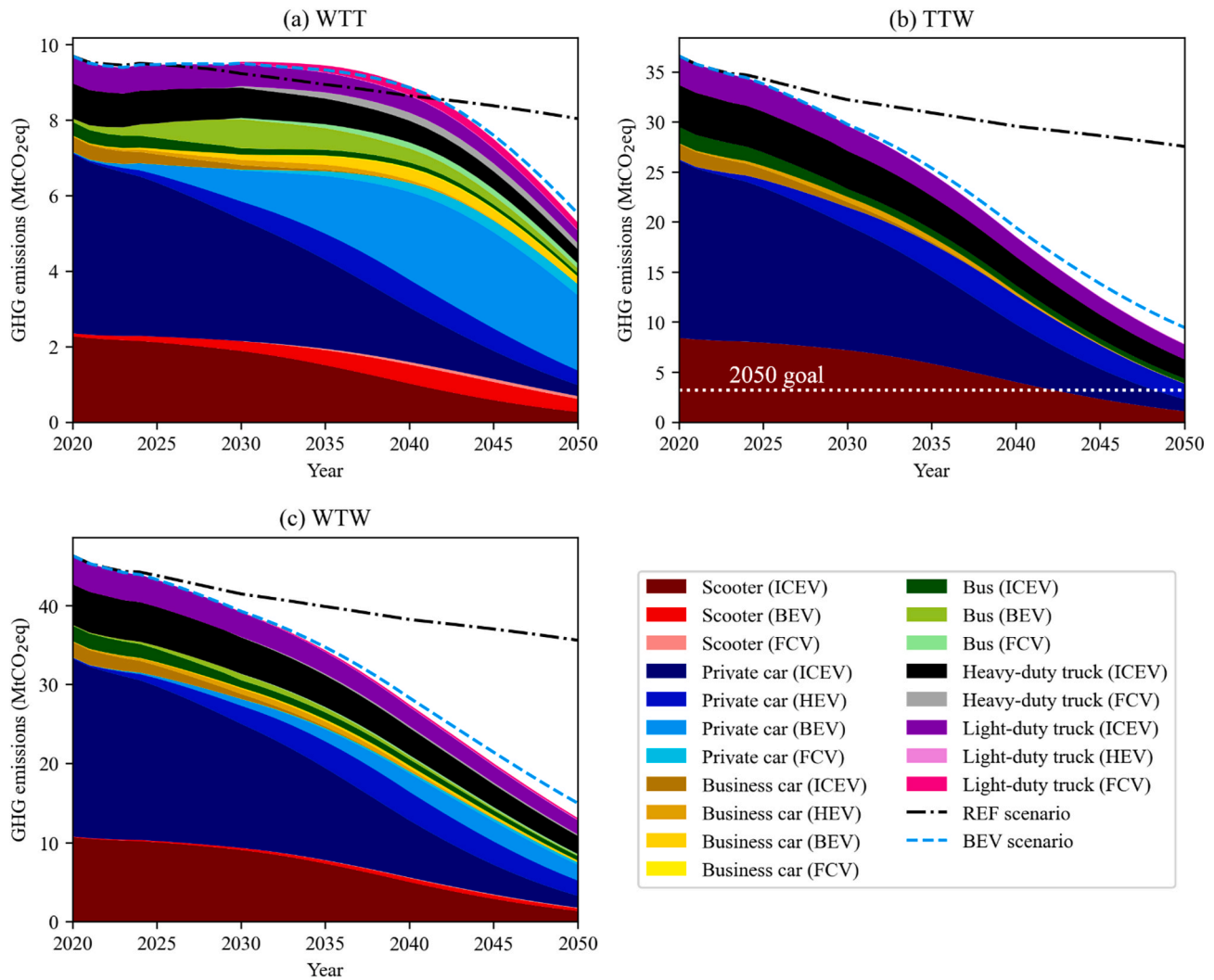


Fig. 9. Scenario analysis results of GHG emissions across (a) WTT, (b) TTW, and (c) WTW stages for on-road vehicles in Taiwan (2020–2050).

vehicles. By examining the WTW stage as a whole, the strategic shift of emissions from direct vehicle operation to upstream processes emerges as a long-term benefit for decarbonization.

Despite expected reductions in fuel life cycle GHG emissions under both the BEV and BEV-FCV scenarios, projections indicate that direct TTW emissions will still surpass Taiwan's 2050 target of 3.19 MtCO₂eq. The current governmental measures, led primarily by scooters and private cars in the BEV scenario, do not fully meet the net-zero goal. The absence of an inclusive electrification strategy for heavier vehicles such as buses and trucks highlights significant opportunities for further

emission reductions. Incorporating hydrogen vehicles into the fleet could potentially reduce TTW emissions to 7.83 MtCO₂eq (95% CI: 6.33–8.41 MtCO₂eq), although this still exceeds the target. This suggests that a more aggressive vehicle electrification strategy is crucial to achieve deeper emission reductions.

3.7. Pathways to emission targets via vehicle fleet composition changes

Table 1 illustrates the anticipated vehicle stock and corresponding TTW GHG emissions by 2050, emphasizing differences in ZEV

Table 1
Projected vehicle stock, TTW GHG emissions, and ZEVs penetration in 2050.

Vehicle type	Vehicle stock prediction in 2050	REF scenario		BEV scenario		BEV-FCV scenario	
		TTW GHG emission (MtCO ₂ eq)	ZEVs stock share (%)	TTW GHG emission (MtCO ₂ eq)	ZEVs stock share (%)	TTW GHG emission (MtCO ₂ eq)	ZEVs stock share (%)
Scooter	12,287,417	6.20	11.94	1.10	84.39	1.10	84.39
Private car	7,322,609	13.97	4.19	2.75	78.66	2.75	78.66
Business car	253,547	1.22	6.05	0.04	96.53	0.04	96.53
Urban bus	17,902	0.45	49.20	0	100	0	100
Tour bus	19,993	0.79	0.83	0.76	0.83	0.46	39.33
HDT	128,443	2.73	0	2.73	0	1.94	28.98
LDT	911,737	2.20	0	2.08	0	1.55	35.37
Total	20,941,586	27.55	8.58	9.45	78.28	7.83	79.63

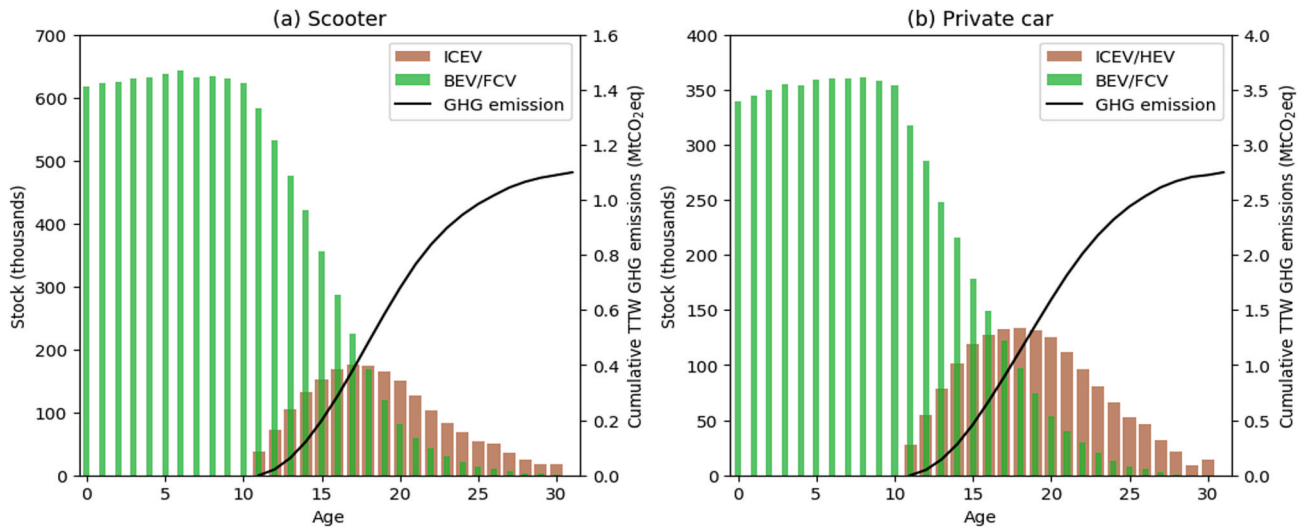


Fig. 10. Projected stocks of scooters and private cars by age and powertrain type, with cumulative TTW GHG emissions under the BEV or BEV-FCV scenario in 2050.

penetration across scenarios and underscoring the need for precise policy measures to meet Taiwan's decarbonization goals. Although HDVs only make up 5.15% of the total vehicle fleet, their disproportionate emission contribution of 3.95 MtCO₂eq, or 50.4% of the sector's emissions, highlights a significant gap in current strategies. Fully incorporating HDVs into the ZEV framework could drastically cut TTW

emissions, compensating for the slower phase-out of ICEVs and HEVs.

LDVs, including scooters and cars, are significantly impacted by existing electrification policies, yet these measures are still insufficient to achieve the 2050 emission targets. In scenarios involving both BEVs and FCVs, LDVs are responsible for 49.6% of transport emissions, with private cars as primary contributors. The transition to ZEVs, particularly

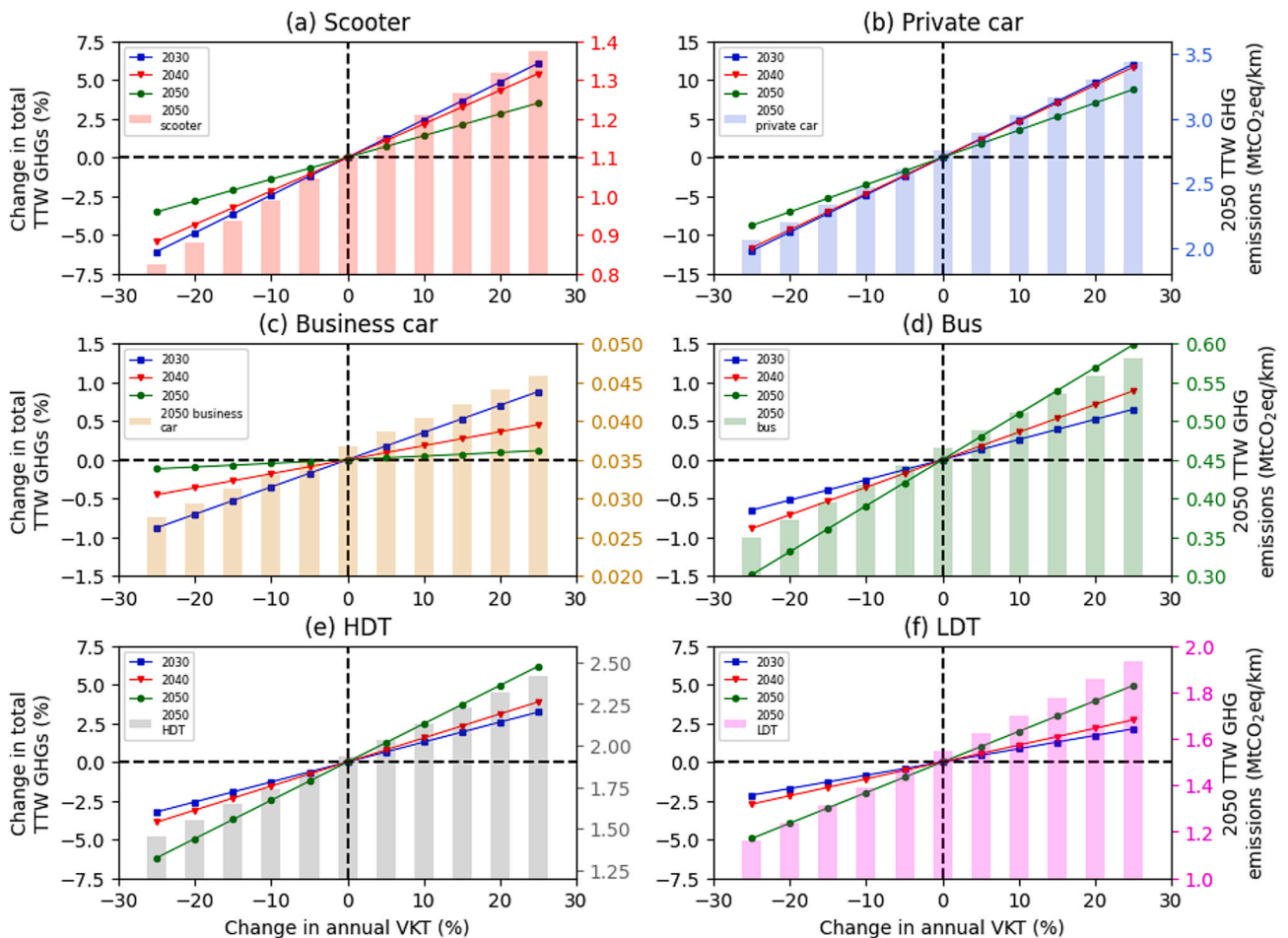


Fig. 11. Sensitivity analysis depicting the potential impact of varying VKT on TTW GHG emissions in 2050 for (a) Scooters; (b) Private cars; (c) Business cars; (d) Buses; (e) HDT; (f) LDT.

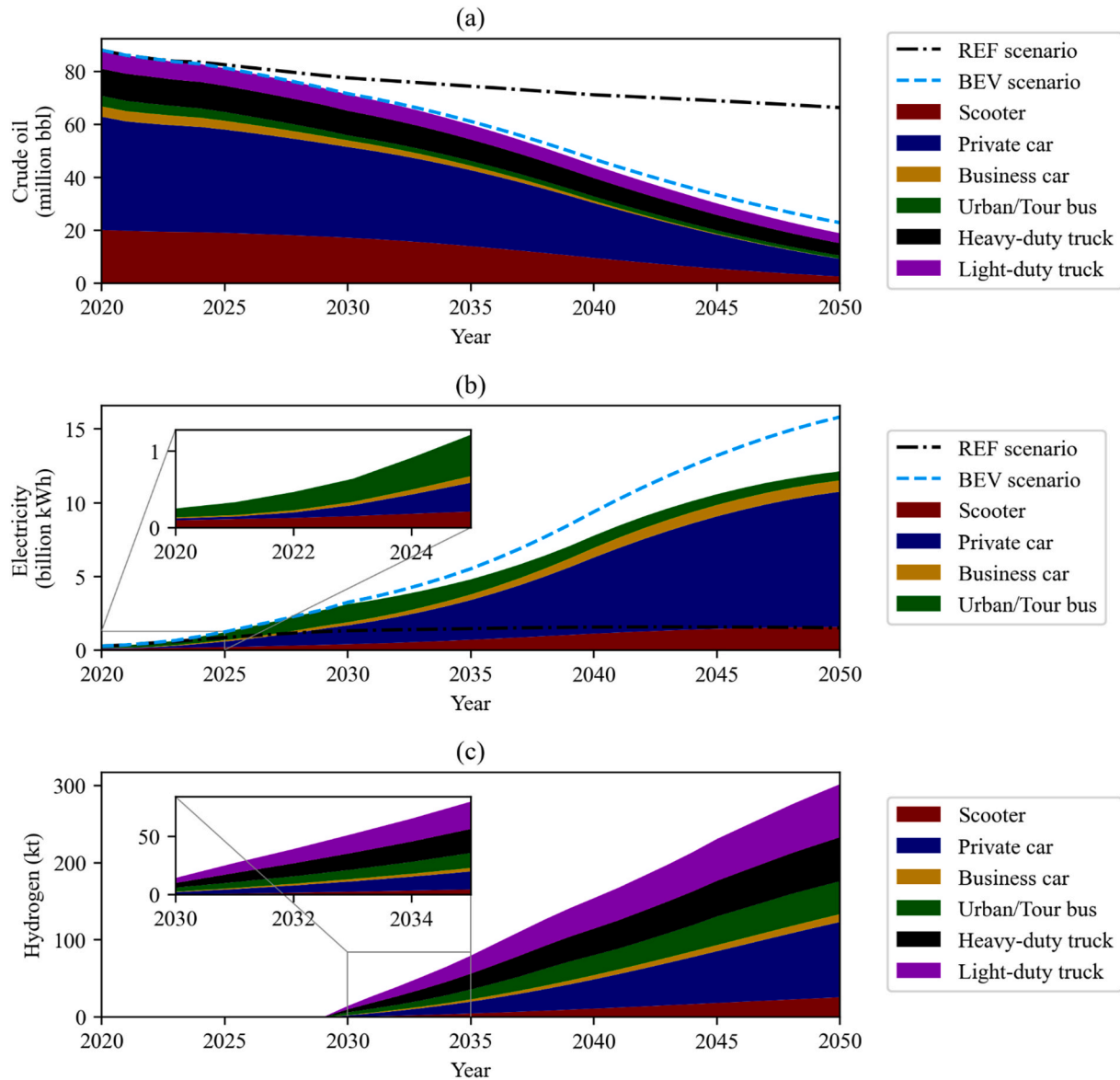


Fig. 12. Projected energy demand for on-road vehicles under the BEV-FCV scenario from 2020 to 2050: (a) Crude Oil, (b) Electricity, and (c) Hydrogen.

for private cars, is hampered by long vehicle replacement cycles. According to Fig. S10, the median lifespan of scooters and cars reaches 17.4 and 20.3 years, respectively, longer than that of business cars, allowing quicker turnover to more efficient models. By 2050, the majority of scooters and cars are expected to shift to ZEVs, especially following projected bans on sales of fossil-fuel vehicles by 2040. However, a residual population of older ICEVs and HEVs will persist, adding to the emissions burden. Fig. 10 outlines the cumulative GHG emissions from these older vehicles, showcasing the significant challenge posed by the remaining fossil-fuel-dependent fleet. To effectively reach the net-zero goal by mid-century, new policies must facilitate the phasing out of older LDVs. Initiatives could include incentives for ZEV upgrades or regulatory measures restricting older vehicle use, accelerating the transition to a greener vehicle fleet.

3.8. Impact of vehicle usage on emission projections

Within the TW-FLEET model, projections of vehicle stocks, sales, and fuel efficiency—which dictate the total emissions and energy consumption of on-road vehicles—are derived from government socio-economic forecasts and academic research. However, the model

assumes a constant figure for VKT based on regulatory inspection data and calculations from the [60]. This constant VKT assumption does not account for potential changes due to varying operational costs, travel purposes, enhancements in public transport, advancements in mobility technology, or evolving transportation policies. To assess VKT's influence on the PTW stage GHG emissions, sensitivity analysis is employed rather than attempting to forecast future VKT trends.

Fig. 11 illustrates the reaction of total GHG emissions for different vehicle types under the BEV-FCV scenario to changes in VKT. By 2050, most LDVs (scooters and cars) are expected to be ZEVs (Table 1), mitigating the impact of VKT increases on direct emissions. However, HDVs (buses and trucks), which represent a significant portion of total emissions, are more affected by VKT fluctuations. Given the uncertainties surrounding future VKT trends, solely relying on modifications in VKT as an emissions reduction method is impractical. Instead, a stronger focus on increasing ZEV market share is advocated, particularly for heavy-duty vehicles. This strategy is crucial for achieving Taiwan's stringent 2050 emissions target of 3.19 MtCO₂eq for the road transport sector.

3.9. Energy shifts in on-road transport

Fig. 12 outlines the transformation of fuel demand in the road transport sector, reflecting the increasing embrace of ZEVs, as delineated in the BEV-FCV scenario. The graph illustrates a significant reduction in traditional fossil fuel use, alongside a marked rise in the demand for electricity and hydrogen, which are pivotal to powering this new fleet. By 2050, FCVs are expected to require about 302 thousand tonnes of hydrogen annually (95% CI: 245.7–324.8 kt), with HDVs consuming over half (58.2%) of this total. Concurrently, the electricity demand for BEVs is projected to jump dramatically, from 0.468 billion kWh in 2022 to 12.14 billion kWh by 2050—a nearly 26-fold increase. This surge contrasts sharply with the anticipated steep decline in crude oil demand—from 84.84 million barrels in 2022 to just 18.90 million barrels in 2050.

The results underscore the imperative for substantial reforms within the energy sector to support a landscape increasingly dominated by electric and hydrogen vehicles. Interestingly, without FCVs in the LDV fleet (as hypothesized in the BEV-only scenario), electricity demand might even reach 15.80 billion kWh by 2050—34 times the 2022 level. Hence, integrating hydrogen vehicles could potentially lessen the burgeoning demand on the electric grid driven by BEVs. However, the challenge of ensuring access to hydrogen—particularly from low-carbon sources—remains critical and will significantly influence the sustainability of the sector moving forward.

4. Conclusion

Building on the research by Tsai et al. [27], this study applies advanced modeling to assess the environmental impact of Taiwan's road transportation sector, with a new focus on integrating hydrogen FCVs into net-zero strategies—a vital consideration absent in current government plans. Utilizing the GREET model tailored with localized data, the LCA scrutinizes hydrogen fuel cycles across the WTG, GTT, and TTW stages. The detailed examination of twenty hydrogen generation processes and ten transportation methods reveals substantial energy requirements for hydrogen's pressurization, liquefaction, or chemical conversion for FCV use. Key findings show that electrolytic hydrogen, especially when generated from renewable energy sources, exhibits the lowest energy consumption and GHG emissions, assuming infrastructural viability. While FCVs currently have higher GHG emissions than ICEVs, projections suggest a future where FCVs could outperform BEVs in environmental efficiency, driven by an increase in green hydrogen production and improvements in vehicle fuel efficiency.

The TW-FLEET model, updated with the latest socio-economic data, forecasts the mix of vehicle types and powertrain technologies and their fuel demands, providing a comprehensive view of future GHG emissions and energy consumption in the transport sector. Results indicate that while BEVs are currently emphasized in policy measures, integrating FCVs could substantially reduce emissions. Notably, HDVs like buses and trucks, though a small proportion of the vehicle fleet by 2050, are expected to account for 50.4% of direct TTW emissions due to their high usage. Converting these to ZEVs could significantly cut emissions by 3.95 MtCO₂eq, underscoring the critical role of HDVs in decarbonization efforts. Additionally, the extended service life of LDVs, such as scooters and private cars, poses challenges to achieving the net-zero goal.

The insights from this study offer valuable guidance for policymakers and industry stakeholders, stressing the need for a comprehensive vehicle electrification strategy that includes FCVs to meet Taiwan's ambitious decarbonization targets. Despite the inherent limitations due to the restricted availability of direct projected data on hydrogen industries in Taiwan, the LCA leverages extrapolations from existing literature and localized databases to provide a forward-looking view of potential environmental impacts. These limitations notwithstanding, the TW-FLEET model is robust, designed to be both flexible and updatable, reflecting shifts in socio-economic indicators, policy

directions, and technological innovations in vehicle types. Moreover, its adaptable framework allows for potential application to the road transport sectors of other countries or regions, thereby offering a scalable tool for global evaluations of vehicular environmental impacts.

CRediT authorship contribution statement

Wei-Hsuan Chen: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Ian Chen:** Writing – review & editing, Visualization, Validation, Investigation. **Jo-Chen Meg Lin:** Writing – review & editing, Project administration, Funding acquisition, Data curation. **I-Yun Lisa Hsieh:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2024.07.234>.

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